

Quantum Mechanics and the Arithmetic Structure of the Bost–Connes System

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Abstract

Starting from the arithmetic structure of the Bost–Connes (BC) system, we systematically investigate the arithmetic origins of each of the five von Neumann axioms of quantum mechanics. The main results are as follows: (I) The Hilbert space structure is uniquely constructed from C^* -positivity; (II) Self-adjointness is an intrinsic feature of the C^* -algebraic framework; (III) The numerical form of the Born rule $P = |c_n|^2$ is uniquely determined by the overlap relation between KMS states; (IV) Unitary evolution is uniquely determined by the strict monotonicity of ζ ; (V) The tensor product decomposition is a direct consequence of the fundamental theorem of arithmetic.

The derivation introduces no measurement, observer, or wave function collapse. Probability emerges naturally within the BC framework as an objective relational property between two KMS states.

1 Introduction

Von Neumann’s axiomatization of quantum mechanics [2] comprises five axioms: (I) the state space is a Hilbert space; (II) observables are self-adjoint operators; (III) measurement probabilities are given by the Born rule; (IV) time evolution is a unitary group; (V) the state space of a composite system is the tensor product. In the standard formulation, these five axioms are regarded as mutually independent physical postulates. In particular, Axiom III (the Born rule) is usually stated as a postulate about *measurement outcomes*—requiring the introduction of observers, measurement apparatus, and wave function collapse.

In this paper, we re-examine these five axioms from the perspective of the Bost–Connes (BC) system [1]. The essential content of all five axioms can be traced to arithmetic origins within the BC system. No measurement, observer, or collapse is introduced. The key insight is that, for $\beta > 1$, the BC system possesses a Galois-transitive family of KMS states $\{\varphi_{\beta,\alpha}\}_{\alpha \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})}$. Different KMS states agree on diagonal quantities (the energy eigenvalues $n^{-\beta}/\zeta(\beta)$) but differ on off-diagonal quantities (phases). The Born rule emerges as the *unique overlap functional* between two such states: probability is not a property of a particle being “measured,” but a *relational property*—the extent to which two KMS states fail to see each other’s phases.

The derivation proceeds in five steps:

1. The positivity of C^* -algebra states uniquely determines the inner product ($p = 2$, not an arbitrary choice);
2. The partition function $Z(\beta) = \zeta(\beta)$ determines the normalization $w_n = 1$;
3. The Galois phase mismatch between KMS states yields the Born rule $P = |c_n|^2$;
4. The strict monotonicity of ζ determines the unique unitary group;
5. The fundamental theorem of arithmetic yields the tensor product.

All structural elements of the BC system are uniquely determined by the arithmetic of $\mathbb{Z}$ (Remark 2).

2 The Bost–Connes system

The scalar field over which the BC system is defined is uniquely determined by arithmetic. Completion (from the rationals to the reals and then to the complex numbers) is the only step in the derivation that introduces continuity—it is a standard mathematical tool required for the GNS construction, not a physical postulate.

Definition 1 (The BC system). *The BC system is defined by the following data [1], divided into arithmetic input and derived structure:*

Arithmetic input (three items):

- (i) *The set of objects $\mathbb{N}^+ = \{1, 2, 3, \dots\}$.*
- (ii) *Unique factorization: $n = \prod_p p^{a_p(n)}$ (fundamental theorem of arithmetic).*
- (iii) *The Hamiltonian $H(n) = \log n$ (the unique nonnegative completely additive function on $\mathbb{N}^+$; Proposition 3).*

Derived structure (four items, each uniquely determined by (i)–(iii)):

- (iv) *Multiplication operators $\mu_m(n) = mn$ (a faithful representation of the multiplicative action of $\mathbb{N}^+$; uniquely determined by (ii)).*
- (v) *The one-parameter automorphism group $\sigma_t(\mu_m) = m^{it}\mu_m$, $t \in \mathbb{R}$ (pointwise multiplication n^{it} on the function space over $\mathbb{N}^+$; uniquely determined by (iii); see Remark 2).*
- (vi) *The partition function $Z(\beta) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta)$, $\beta > 1$ ($= \sum_n n^{-\beta}$ by direct summation; uniquely determined by (iii)).*
- (vii) *KMS states ($\beta > 1$): $\varphi_\beta(|n\rangle\langle m|) = \delta_{nm} n^{-\beta}/\zeta(\beta)$ (uniquely determined by the KMS condition for the dynamics σ_t ; this is the main theorem of [1]).*

Remark 2 (Uniqueness of the derived structure). *Items (iv)–(vii) are not independent assumptions; each is a unique mathematical consequence of (i)–(iii):*

- (iv): *The multiplicative action $m \cdot n = mn$ is the unique faithful action of $\mathbb{N}^+$ on itself compatible with unique factorization.*
- (v): *The map $\sigma_t(f)(n) = n^{it}f(n)$ is pointwise multiplication on the function space over $\mathbb{N}^+$ —a purely algebraic operation requiring no Hilbert space. Its uniqueness as a one-parameter automorphism group is established in Theorem 12 (Section 6).*
- (vi): *$Z(\beta) = \sum_n e^{-\beta \log n} = \sum_n n^{-\beta}$ is a direct summation from H ; no trace or inner product is needed.*
- (vii): *KMS states at inverse temperature β are uniquely determined by the dynamics σ_t (unique for $\beta < 1$; a Galois-transitive family for $\beta > 1$ [1]).*

The BC algebra can be defined abstractly via generators μ_n , $e(r)$ ($r \in \mathbb{Q}/\mathbb{Z}$) and relations [1]; the concrete representation on $\ell^2(\mathbb{N}^+)$ is an output of the GNS construction (Theorem 4), not an input. The C^* -algebraic language is a mathematical framework, not a physical postulate. The physical input consists solely of items (i)–(iii), namely the arithmetic of $\mathbb{Z}$.

Proposition 3 (Uniqueness of H). *Let $f : \mathbb{N}^+ \rightarrow \mathbb{R}$ satisfy (a) $f(mn) = f(m) + f(n)$, (b) $f(n) \geq 0$, (c) $f(n+1) - f(n) \rightarrow 0$. Then $f = c \log n$ for some $c \geq 0$.*

Proof. This follows from the Erdős–Kátaı theorem [4, 5]. An independent proof uses Ostrowski’s theorem: $e^{f(n)}$ is an absolute value on $\mathbb{Z}$; the condition $f \geq 0$ excludes all p -adic absolute values, leaving only the Archimedean absolute value n^c [6]. $\square$

3 Axiom I: Hilbert space

Theorem 4. *The C^* -algebraic positivity of KMS states uniquely determines the inner product $\langle m|n\rangle = \delta_{mn}$ and the state space $\ell^2(\mathbb{N}^+)$.*

Proof. No inner product is assumed a priori. The construction proceeds in four steps from the BC data.

Step 1 (Positivity $\Rightarrow$ sesquilinear form). Each KMS state φ_β is a positive linear functional on the BC algebra $\mathcal{A}$: $\varphi_\beta(a^*a) \geq 0$ for all $a \in \mathcal{A}$. This positivity defines a sesquilinear form on $\mathcal{A}$:

$$\langle a, b \rangle_\varphi = \varphi_\beta(b^*a). \quad (1)$$

This form automatically possesses conjugate symmetry ($\langle a, b \rangle_\varphi = \overline{\langle b, a \rangle_\varphi}$) and positive semi-definiteness ($\langle a, a \rangle_\varphi \geq 0$). This is a mathematical consequence of C^* -positivity—not a physical postulate.

Why $p = 2$: $\varphi(b^*a)$ is a sesquilinear form—that is, a map simultaneously linear (antilinear) in a and b . A sesquilinear form is by definition an inner product—and the completion of an inner product space is L^2 (a Hilbert space). The exponent $p = 2$ is not a choice; it is the intrinsic structure of a sesquilinear form—namely, the positivity $\varphi(a^*a) \geq 0$ of C^* -algebra states.

Step 2 (Quotient and completion $\Rightarrow$ Hilbert space). Taking the quotient by the null space $\{a : \langle a, a \rangle_\varphi = 0\}$ and completing yields the Hilbert space $\mathcal{H}_\varphi$. For the BC system, the KMS states at $\beta > 1$ satisfy $\varphi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta) > 0$ for all n , so no nonzero basis vector is null. The resulting Hilbert space is $\mathcal{H}_\varphi = \ell^2(\mathbb{N}^+)$.

Step 3 (Orthogonality). Under the GNS inner product $\langle a, b \rangle_\varphi = \varphi_\beta(b^*a)$, H as an element of the BC algebra satisfies the symmetry condition: $\varphi_\beta((Hg)^*f) = \varphi_\beta(g^*Hf)$, since H is self-adjoint ($H^* = H$, $\log n \in \mathbb{R}$). In the GNS representation, this is equivalent to:

$$\langle Hf, g \rangle_\varphi = \sum_n (\log n) \overline{f(n)} g(n) = \langle f, Hg \rangle_\varphi. \quad (2)$$

Since H has simple spectrum ($\log m \neq \log n$ for $m \neq n$),

$$(\log m - \log n) \langle m|n \rangle_\varphi = 0 \implies \langle m|n \rangle_\varphi = 0 \quad (m \neq n). \quad (3)$$

Step 4 (Normalization $w_n = 1$). The GNS construction yields the existence of the inner product and orthogonality (Steps 1–3), but the norm of each basis vector is not yet determined. The GNS inner product gives $\langle n|n \rangle_\varphi = \varphi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta)$, which is not the standard normalization. The standard normalization is obtained by rescaling: define $|\tilde{n}\rangle = \sqrt{\zeta(\beta)/n^{-\beta}} |n\rangle$, so that $\langle \tilde{m}|\tilde{n} \rangle_\varphi = \delta_{mn}$.

This rescaling is not a convention—it is uniquely determined by the partition function. In the standard basis $\{|\tilde{n}\rangle\}$, the partition function is $Z(\beta) = \sum_n \langle \tilde{n}|e^{-\beta H}|\tilde{n} \rangle_\varphi = \sum_n n^{-\beta} = \zeta(\beta)$, with coefficients exactly equal to 1. By the uniqueness of Dirichlet series coefficients [18], this is the unique normalization satisfying $Z = \zeta$.

In summary: GNS yields the structure (orthogonality), while $Z = \zeta$ yields the normalization ($w_n = 1$). The two are compatible but must be treated separately.

Conclusion. $\langle m|n \rangle = \delta_{mn}$. The state space is $\ell^2(\mathbb{N}^+)$ —constructed, not assumed a priori. $\square$

Remark 5. *The above construction is the GNS construction applied to the BC partition state. We emphasize that the GNS construction is not an external tool imported from C^* -algebra theory—it is the unique mathematical consequence of positivity $\varphi(a^*a) \geq 0$, which is itself a consequence of the BC algebra being uniquely determined by $\mathbb{Z}$ (Remark 2).*

The state space $\ell^2(\mathbb{N}^+)$ is not assumed a priori—it uniquely emerges from the positivity of KMS states of the BC algebra; the discrete physical space gives rise to a continuous Hilbert structure through completion.

4 Axiom II: Self-adjoint observables

Theorem 6. *The observables of the BC system—operators that commute with H and are generated by arithmetic functions—are all self-adjoint.*

Proof. Self-adjointness of H . The domain is $\text{dom}(H) = \{f \in \ell^2 : \sum_n (\log n)^2 |f(n)|^2 < \infty\}$. Symmetry: for $f, g \in \text{dom}(H)$, $\langle Hf, g \rangle = \langle f, Hg \rangle$. The set $\{|n\rangle\}$ constitutes a complete eigenbasis with real eigenvalues $\log n$; a symmetric operator with a complete real eigenbasis is self-adjoint (deficiency indices $(0, 0)$) [7].

Simple spectrum. $\log m = \log n \Leftrightarrow m = n$ ($\log$ is injective on $\mathbb{N}^+$), so every eigenspace of H is one-dimensional.

Commutant. If a bounded operator A commutes with H , then simplicity of the spectrum forces A to preserve each one-dimensional eigenspace, so $A|n\rangle = a_n|n\rangle$ (i.e., A must be diagonal).

Arithmetic observables are self-adjoint. The diagonal operators naturally arising from the BC system include: $H(n) = \log n$ (energy, $\log n \in \mathbb{R}_{\geq 0}$), $\hat{n}_p(n) = v_p(n)$ (p -channel occupation number, $v_p \in \mathbb{Z}_{\geq 0}$), $\hat{N}(n) = \Omega(n) = \sum_p v_p(n)$ (total number of prime factors, $\Omega \in \mathbb{Z}_{\geq 0}$), and $\mu_m \mu_m^*$ (projections, eigenvalues $\in \{0, 1\}$). All of these operators have nonnegative real eigenvalues—this is not an assumption, but a direct consequence of the ranges of the arithmetic functions $\log$, v_p , and Ω . A diagonal operator with real eigenvalues is self-adjoint. $\square$

Remark 7. *In the C^* -algebraic framework, observables are defined as the self-adjoint part of the algebra $\mathcal{A}_{\text{sa}} = \{a \in \mathcal{A} : a^* = a\}$. This is not an additional physical postulate—it is an intrinsic feature of C^* -algebras, and the BC algebra is uniquely determined by the arithmetic of $\mathbb{Z}$ (Remark 2). The operators naturally arising from the BC system—namely those defined by arithmetic functions—have intrinsically real eigenvalues ($\log n, v_p(n), \Omega(n) \in \mathbb{R}_{\geq 0}$), and hence belong to $\mathcal{A}_{\text{sa}}$. Therefore, Axiom II is not an independent postulate within the BC framework, but a constituent part of the C^* -algebra definition. Arithmetic functions on $\mathbb{Z}$ are inherently real-valued; self-adjointness is a consequence of arithmetic.*

5 Axiom III: The Born rule

Theorem 8. *The overlap between distinct KMS states of the BC algebra uniquely determines the Born rule $P(n|\psi) = |\langle n|\psi \rangle|^2$.*

Proof. This proof derives the Born rule from the relational structure of KMS states—without introducing measurement, observers, or collapse.

Step 1 (KMS states as relational objects). For $\beta > 1$, the BC system possesses a family of KMS states $\{\varphi_{\beta, \alpha}\}_{\alpha \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})}$, related by the Galois action [1]. All states agree on diagonal quantities:

$$\varphi_{\beta, \alpha}(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta) \quad (\text{independent of } \alpha). \quad (4)$$

They differ on off-diagonal quantities (phases): $\varphi_{\beta, \alpha}(\zeta_n) = \alpha(\zeta_n)$.

Step 2 (Orthogonality $\rightarrow$ functional form of probability). Let $|\psi\rangle = \sum_n c_n |n\rangle$ be a state in $\ell^2(\mathbb{N}^+)$. By orthogonality (Theorem 4, Step 3),

$$\langle n|\psi \rangle = \sum_m c_m \langle n|m \rangle = \sum_m c_m \delta_{nm} = c_n. \quad (5)$$

Orthogonality eliminates all terms with $m \neq n$. Therefore, the *entire information* of $|\psi\rangle$ in the $|n\rangle$ direction is c_n —no other coefficient $\{c_m\}_{m \neq n}$ can influence this projection.

Now let $P(n|\psi)$ denote the probability of $|\psi\rangle$ in the $|n\rangle$ direction. By the above argument, $P(n|\psi)$ can depend only on c_n (since this is the sole information of $|\psi\rangle$ in the $|n\rangle$ direction).

Global phase invariance ($|\langle n | e^{i\theta} \psi \rangle| = |c_n|$, since probabilities should not depend on the choice of global phase of the state vector) further excludes dependence on the phase of c_n . Therefore:

$$P(n|\psi) = f(|c_n|). \quad (6)$$

No Riesz representation theorem is used here—the argument relies entirely on orthogonality and phase invariance.

Since H has simple spectrum (log is injective on $\mathbb{N}^+$), $\{|n\rangle\}$ is the unique eigenbasis. Non-contextuality holds automatically: simple spectrum $\rightarrow$ eigenbasis $\{|n\rangle\}$ unique $\rightarrow$ no freedom to choose a different basis $\rightarrow$ non-contextuality.

That f is independent of n follows from normalization. First consider the two-component case: for any m, n, n' , take $|\psi\rangle = \alpha|m\rangle + \beta|n\rangle$ and $|\psi'\rangle = \alpha|m\rangle + \beta|n'\rangle$; normalization gives $f(|\alpha|) + f_n(|\beta|) = 1 = f(|\alpha|) + f_{n'}(|\beta|)$, hence $f_n = f_{n'}$. The general case follows by induction: assume f is independent of n for states with k components. Take a $(k+1)$ -component state $|\psi\rangle = \alpha|m\rangle + \beta|\psi_{\text{rest}}\rangle$, where $|\psi_{\text{rest}}\rangle$ has k components. Normalization gives $f(|\alpha|) + \sum_j f(|c_{n_j}|) = 1$; by the inductive hypothesis, $\sum_j f(|c_{n_j}|)$ is independent of the specific choice of $\{n_j\}$, so $f(|\alpha|)$ is also independent of m . The induction is complete.

Step 4 (C^* -positivity constraints on f). The function $f : [0, 1] \rightarrow [0, 1]$ satisfies:

- (a) $f(x) \geq 0$ (C^* -state positivity: $\varphi(a^*a) \geq 0$);
- (b) $\sum_n f(|c_n|) = 1$ (normalization: $\varphi_\beta(I) = 1$);
- (c) $f(0) = 0$ (orthogonality: $\langle m|n\rangle = 0$ for $m \neq n$, Theorem 4);
- (d) $\sum_{n \in A} f(|c_n|) = f_A$ for mutually exclusive outcomes ($\mathbb{N}^+$ is discrete; on a discrete space, $P(A) = \sum_{n \in A} P(n)$ is the definition of event probability).

Step 5 (Determining f on a dense set—using only eigenstates). For equal-amplitude states $|\psi\rangle = \frac{1}{\sqrt{k}} \sum_{j=1}^k |n_j\rangle$, condition (b) gives $f(1/\sqrt{k}) = 1/k$.

Now take j basis states with amplitude $x_0 = 1/\sqrt{k}$ and one remainder state:

$$|\psi\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^j |n_i\rangle + \sqrt{1 - j/k} |n_{j+1}\rangle, \quad (7)$$

already normalized: $j/k + (1 - j/k) = 1$. By (b):

$$j \cdot f(1/\sqrt{k}) + f(\sqrt{1 - j/k}) = 1 \implies f(\sqrt{m/k}) = m/k, \quad (8)$$

where $m = k - j$ ranges over $0, 1, \dots, k$. The set $\{\sqrt{m/k} : 0 \leq m \leq k, k \in \mathbb{N}^+\}$ is dense in $[0, 1]$. On this dense set, $f(x) = x^2$.

No coarse-graining, no change of basis, no subspace projections. The entire argument uses only the eigenstates $\{|n\rangle\}$ and the normalization condition (b).

Step 6 (Cauchy equation + non-negativity $\Rightarrow f(a) = a^2$ for all a). Define $h(t) = f(\sqrt{t})$ for $t \in [0, 1]$, so that $f(x) = h(x^2)$.

Cauchy equation. Take three distinct eigenstates $|1\rangle, |2\rangle, |3\rangle$ and form two states:

$$|\psi_1\rangle = \sqrt{s}|1\rangle + \sqrt{t}|2\rangle + \sqrt{1-s-t}|3\rangle, \quad (9)$$

$$|\psi_2\rangle = \sqrt{s+t}|1\rangle + \sqrt{1-s-t}|2\rangle, \quad (10)$$

where $s, t \geq 0$ and $s + t \leq 1$. Both are finite-component normalized states. Applying condition (b) to each:

$$h(s) + h(t) + h(1-s-t) = 1, \quad (11)$$

$$h(s+t) + h(1-s-t) = 1. \quad (12)$$

Subtracting:

$$h(s+t) = h(s) + h(t) \quad (s, t \geq 0, s+t \leq 1). \quad (13)$$

This is Cauchy's functional equation on $[0, 1]$. Only finite-component states are used; no σ -additivity is needed.

Solution on $\mathbb{Q}$. By Step 5, $f(\sqrt{m/k}) = m/k$, i.e., $h(m/k) = m/k$ for all rationals $m/k \in [0, 1]$. Thus h equals the identity on $\mathbb{Q} \cap [0, 1]$.

Extension to all of $[0, 1]$ via non-negativity. For arbitrary $\alpha \in [0, 1]$, possibly irrational:

Lower bound. For any rational r with $0 \leq r < \alpha$: $h(\alpha) = h(r) + h(\alpha - r) = r + h(\alpha - r) \geq r$, since $h(\alpha - r) \geq 0$ (condition (a)). Taking the supremum over all such r : $h(\alpha) \geq \alpha$.

Upper bound. For any rational r with $\alpha < r \leq 1$: $h(r) = h(\alpha) + h(r - \alpha)$, so $r = h(\alpha) + h(r - \alpha)$, giving $h(\alpha) = r - h(r - \alpha) \leq r$, since $h(r - \alpha) \geq 0$ (condition (a)). Taking the infimum over all such r : $h(\alpha) \leq \alpha$.

Combining: $h(\alpha) = \alpha$ for all $\alpha \in [0, 1]$.

Conclusion. $f(x) = h(x^2) = x^2$ for all $x \in [0, 1]$. Therefore $P(n|\psi) = |c_n|^2 = |\langle n|\psi \rangle|^2$.

Summary of inputs: the Cauchy equation uses only finite-component normalization (b); the extension uses only non-negativity (a) (C^* -positivity) and the density of $\mathbb{Q}$ in $\mathbb{R}$. No monotonicity, continuity, σ -additivity, or infinite-dimensional states are required. $\square$

Remark 9 (Origin of all conditions within the BC system). • **(a) Positivity:** $\varphi(a^*a) \geq 0$ is the defining property of C^* -states, uniquely determined by $\mathbb{Z}$ (Remark 2).

- **(b) Normalization:** $\varphi_\beta(I) = \zeta(\beta)/\zeta(\beta) = 1$.
- **(c) Orthogonality:** Theorem 4, Step 3.
- **(d) Monotonicity:** The inner product is the unique GNS inner product; it defines a metric on the state space; the overlap in a metric space is constructively monotone.
- **(e) Additivity:** Discrete sample space $\mathbb{N}^+$; $P(A) = \sum_{n \in A} P(n)$ is the definition.
- **Non-contextuality:** Orthogonality $\langle m|n \rangle = 0$ eliminates all $m \neq n$ terms in $\langle n|\psi \rangle = c_n$ (Step 3).
- **f independent of n :** Normalization (Step 3).
- **No basis-independence required:** Step 5 uses only the eigenstates $\{|n\rangle\}$ and normalization—no coarse-graining.

Remark 10 (Kolmogorov axioms). The Kolmogorov axioms (K1) non-negativity, (K2) normalization, (K3) σ -additivity are all consequences of the BC structure: (K1) from C^* -positivity; (K2) from $\zeta(\beta)/\zeta(\beta) = 1$; (K3) from the discreteness of $\mathbb{N}^+$.

Remark 11 (Probability without measurement). In this derivation, probability is not a statement about “what happens upon measurement.” It is the overlap between two states in the GNS Hilbert space—a purely mathematical quantity determined by C^* -positivity. The Born rule $P = |c_n|^2$ answers the question: “From the perspective of the state $|n\rangle$, how much of the state $|\psi\rangle$ is visible?” This is a relational property between two states, not a property of a measurement apparatus.

6 Axiom IV: Unitary time evolution

Theorem 12 (No additional assumptions). The BC system possesses a unique strongly continuous one-parameter unitary group

$$U(t)|n\rangle = e^{-it \log n}|n\rangle, \quad t \in \mathbb{R}, \quad (14)$$

with generator $H = \log n$: $U(t) = e^{-itH}$. The one-parameter automorphism group σ_t and the time parameter t are both uniquely determined by the arithmetic data of the BC system.

Proof. Unitarity. $\|U(t)f\|^2 = \sum_n |e^{-it \log n}|^2 |f(n)|^2 = \|f\|^2$. $U(t)^* = U(-t)$, $U(t)^*U(t) = I$.

Group law. $U(s)U(t)|n\rangle = e^{-i(s+t) \log n}|n\rangle = U(s+t)|n\rangle$. $U(0) = I$.

Strong continuity. $\|U(t)f - f\|^2 = \sum_n |e^{-it \log n} - 1|^2 |f(n)|^2$. For each n : $|e^{-it \log n} - 1| \rightarrow 0$ as $t \rightarrow 0$. The integrand is bounded by $4|f(n)|^2 \in \ell^1$. By the dominated convergence theorem [7], $\|U(t)f - f\| \rightarrow 0$.

Generator. $-i \frac{d}{dt} U(t)|n\rangle|_{t=0} = (\log n)|n\rangle = H|n\rangle$. Therefore $U(t) = e^{-itH}$ (H is self-adjoint, by Theorem 6).

Uniqueness. (The key to the following argument is that the full structure of the BC algebra—including the multiplication operators μ_m and the group algebra elements $e(r)$ —jointly constrains the form of the automorphism group.) Let σ'_t be another continuous one-parameter $*$ -automorphism group on the BC algebra. Any $*$ -automorphism of the BC algebra must preserve its defining relations; in particular, σ'_t must preserve the multiplicative relation $\mu_m \mu_n = \mu_{mn}$. Since σ'_t is a *strongly continuous* one-parameter group with $\sigma'_0 = \text{id}$, and the generators μ_m are indexed by the discrete set $\mathbb{N}^+$, a continuous path from the identity cannot permute them; hence σ'_t must fix each μ_m individually (up to a scalar): $\sigma'_t(\mu_m) = \lambda_m(t) \mu_m$. Since μ_m is an isometry ($\mu_m^* \mu_m = I$), its range projection $\mu_m \mu_m^*$ projects onto multiples of m , and a $*$ -automorphism simultaneously preserves the isometry condition and the lattice of range projections. Therefore, $\sigma'_t(\mu_m) = \lambda_m(t) \mu_m$, where $\lambda_m(t) \in \mathbb{C}$, $|\lambda_m(t)| = 1$ ($*$ -automorphisms preserve norms), $\lambda_{mn}(t) = \lambda_m(t) \lambda_n(t)$ (multiplicative compatibility), $\lambda_m(t+s) = \lambda_m(t) \lambda_m(s)$ (group law).

Multiplicative compatibility implies $\lambda_m(t) = m^{ig(t)}$ (complete multiplicativity + unitarity $\Rightarrow$ exponential form). The group law and continuity give $g(t+s) = g(t) + g(s)$ (Cauchy functional equation); the unique continuous solution is $g(t) = ct$ for some $c \in \mathbb{R}$.

The corresponding Hamiltonian is $H' = cH = c \log n$, with partition function

$$Z'(\beta) = \sum_n n^{-c\beta} = \zeta(c\beta). \quad (15)$$

The BC system requires $Z(\beta) = \zeta(\beta)$. Therefore, $\zeta(c\beta) = \zeta(\beta)$ for all $\beta > 1$. Since ζ is strictly decreasing on $(1, \infty)$ (each term $n^{-\beta}$ is strictly decreasing), $c\beta = \beta$, hence $c = 1$. Thus $\sigma'_t = \sigma_t$. $\square$

Remark 13. *Neither Stone's theorem (the existence of $U(t)$ follows directly from the explicit formula $e^{-it \log n}$) nor the Tomita–Takesaki theorem (uniqueness follows from the monotonicity of ζ) is needed here. “Time” is not introduced as an external physical interpretation—it is the unique intrinsic one-parameter dynamical automorphism group of the BC system, determined by $Z(\beta) = \zeta(\beta)$ and the strict monotonicity of ζ . The strict monotonicity of ζ rules out all alternatives; the existence and uniqueness of time are arithmetic facts.*

7 Axiom V: Tensor product

Theorem 14 (No additional assumptions). *The fundamental theorem of arithmetic yields the unitary isomorphism*

$$\ell^2(\mathbb{N}^+) \cong \bigotimes_{p \text{ prime}} \ell^2(\mathbb{Z}_{\geq 0}), \quad (16)$$

where the right-hand side is the von Neumann infinite tensor product [3] stabilized by $|0\rangle_p$.

Proof. Correspondence of bases. The unique factorization $n = \prod_p p^{a_p}$ gives a bijection $\mathbb{N}^+ \leftrightarrow \bigoplus_p \mathbb{Z}_{\geq 0}$, inducing $|n\rangle \leftrightarrow \bigotimes_p |a_p(n)\rangle_p$. Only finitely many $a_p \neq 0$ ($|0\rangle_p$ serves as the stabilizing vector for almost all p).

Orthogonality. $\langle m|n\rangle = \delta_{mn} \Leftrightarrow \prod_p \delta_{a_p(m), a_p(n)} = \delta_{mn}$.

Completeness. $\{|n\rangle\}$ and $\{\bigotimes_p |a_p\rangle_p\}$ are complete orthonormal bases of their respective spaces. A bijection between complete orthonormal bases extends uniquely to a unitary isomorphism. $\square$

Remark 15. *The Hamiltonian decomposes as $H = \sum_p (\log p) \hat{n}_p$ —the energy sum of independent bosonic channels. The Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ is the manifestation of this tensor product decomposition in the partition function. The correspondence between positive integers and prime-channel Fock states was independently observed by Spector [15] and Julia [16] within the framework of arithmetic quantum field theory. The contribution of the present paper is to place this observation within the complete axiomatic chain derived from the BC system.*

Remark 16. *The stabilizing vector $|0\rangle_p$ corresponds to $n = 1$ (all $a_p = 0$), i.e., the Fock vacuum of the BC system. In rings where unique factorization fails (e.g., $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$), the bijection of bases does not hold, and the tensor product decomposition ceases to exist [3]. We note that Theorem 14 establishes the subsystem decomposition of the BC state space, rather than the general quantum mechanical principle of composing arbitrary systems via tensor products; the latter requires additional physical input.*

Remark 17. *The tensor product structure is not a postulate about system composition—it is a direct expression of the unique factorization $n = \prod p^{a_p}$. The independence of the prime channels is the independence of subsystems.*

8 Discussion

8.1 Summary

Axiom	QM claim	BC determination	Thm
I	Hilbert space	C^* -positivity $\rightarrow$ GNS construction $\rightarrow \ell^2(\mathbb{N}^+)$	4
II	Self-adjoint observables	Self-adjoint part of C^* -algebra + real range of arithmetic functions	6
III	Born rule	KMS overlap + orthogonality + normalization $\rightarrow c_n ^2$	8
IV	Unitary evolution	Strict monotonicity of $\zeta \rightarrow c = 1 \rightarrow \sigma_t$ unique	12
V	Tensor product	Unique factorization $\rightarrow$ prime channel decomposition	14

The above table summarizes the arithmetic origin of each axiom within the BC system. Among these, Axiom IV (uniqueness of unitary evolution) and Axiom V (tensor product decomposition) are strict theorems of the BC system; the essential content of Axioms I and II is naturally obtained from the C^* -algebraic framework; the numerical form of Axiom III, $P = |c_n|^2$, is uniquely determined by the relational structure of KMS states.

8.2 Open problems

1. Is the BC system the unique arithmetic structure capable of producing all five axioms?

9 Conclusion

Within the framework of the BC system, all five von Neumann axioms of quantum mechanics can be traced to the arithmetic structure of $\mathbb{Z}$: the Hilbert space emerges from C^* -positivity, self-adjointness is an intrinsic feature of the algebra, the Born rule is uniquely determined by orthogonality, normalization, and the infinitude of $\mathbb{N}^+$, unitary evolution is locked in by the strict monotonicity of ζ , and the tensor product decomposition is a direct expression of the fundamental theorem of arithmetic.

This result shows that the five axioms of quantum mechanics, within the BC system, are not mutually independent physical postulates, but rather different facets of one and the same arithmetic structure.

Conflict of interest statement

The author declares no conflict of interest.

Data availability statement

This research did not generate or analyze any data. All results are analytical derivations.

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